ML-DL-Ops Assignment 4

Optimizing Transformer Translation with Ray Tune & Optuna

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GitHub | HuggingFace Model

1. Overview

This report documents the hyperparameter optimization of a custom PyTorch Transformer model for English-to-Hindi translation. The baseline model was trained for 100 epochs with fixed hyperparameters on kaggle. Ray Tune paired with the Optuna search algorithm (Bayesian optimization) was then used to find an optimal configuration. The goal was to match or exceed the baseline BLEU score in significantly fewer epochs.

2. Baseline Performance

The baseline was run with a fixed set of hyperparameters: Adam optimizer, `lr = 1e-4`, `batch_size = 32`, `d_model = 512`, `num_heads = 8`, `d_ff = 2048`, `dropout = 0.1`. Each epoch took approximately 66 seconds, giving a total runtime of around 110 minutes.

Training Time	Final Loss	BLEU Score
∼110 minutes	0.0992	71.47
(100 epochs, ∼66 s/ep)	(CrossEntropy)	(NLTK eval)

3. Hyperparameter Search Setup

Ray Tune with **OptunaSearch** and the **ASHA Scheduler** (Asynchronous Successive Halving) were configured to run **20 trials**, each capped at **30 epochs**. ASHA automatically pruned underperforming trials after a grace period of 5 epochs (`reduction_factor = 2`), concentrating compute on the most promising configurations. The total sweep completed in **103.9 minutes**.

The following 7 hyperparameters were included in the search space:

Hyperparameter	Range / Choices	API Method	Rationale
Learning Rate	1e-5 to 1e-2	`loguniform`	Convergence speed
Batch Size	16, 32, 64, 128	`choice`	Gradient noise tradeoff
Attention Heads	4, 8, 16	`choice`	Expressiveness
Feed-Forward Dim	1024, 2048	`choice`	Model capacity
Dropout Rate	0.1 to 0.4	`uniform`	Regularisation
Optimizer	Adam, AdamW	`choice`	Weight decay effect
Weight Decay	1e-5 to 1e-2	`loguniform`	L2 penalty (AdamW)

4. Best Configuration Found

The OptunaSearch algorithm identified the following configuration as optimal after 20 trials:

Parameter	Best Value
Learning Rate	0.000212 (2.12×10^{-4})
Batch Size	16
Attention Heads	16
Feed-Forward Dim	2048
Dropout	0.187
Optimizer	AdamW
Weight Decay	0.00271
Epochs (trial cap)	30
Best Trial Loss	**0.2063**

5. Final Results & Comparison

The best configuration was retrained from scratch for 30 epochs. The table below compares the tuned model against the 100-epoch baseline.

Metric	Baseline (100 ep)	Best Tuned (30 ep)	Improvement
Training Time	~110 min	25.3 min	4.3× faster
Final Loss	0.0992	0.1082	Comparable
BLEU Score	71.47	**90.38**	**+18.91 pts**
Epochs Used	100	30	70% fewer

Key Result: The tuned model achieved a BLEU score of **90.38** versus the baseline **71.47** — an improvement of **+18.91 points** — while using only **30 epochs** (70% fewer) and completing in **25.3 minutes** (4.3× faster). The efficiency goal was decisively exceeded.

6. Analysis & Key Insights

- **Small batch size (16) was decisive.** Smaller batches introduce gradient noise that acts as implicit regularisation, helping the model generalise better and escape local minima faster — critical for a compact translation dataset.

- **AdamW outperformed Adam.** AdamW decouples L2 regularisation from the gradient update. Combined with `weight_decay = 0.00271`, this suppressed overfitting within 30 epochs.

- **16 attention heads improved alignment.** With `d_model = 512` fixed (`head_dim = 32` each), 16 heads allow the model to attend across more diverse subspace representations simultaneously, improving source–target alignment quality.

- **Cosine LR scheduling accelerated convergence.** `CosineAnnealingLR` with `lr = 2.12e-4` enabled rapid early learning followed by smooth fine-tuning, driving loss from 5.12 down to 0.11 in 30 epochs.
- **ASHA scheduler saved compute.** Underperforming trials were pruned after 5 epochs, allowing the full 20-trial sweep to complete in 103.9 minutes on a single GPU — making the search practical.

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